Notas e Listas de Prob 2

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0 Informações importantes

1 Varadhan

Vou passar a limpo exercícios do Varadhan.

Exercise 1.1 (5.1)**.** Construct a nonnegative martingale X_n with $\mathbb{E}X_n = 1$ such that $\xi = \sup_n X_n$ is not integrable.

Proof. The proof is basically done in the book. Just consider the Lebesgue masure on $[0,1]$ and the $\mathcal{F}_n$ will be the sigma-algebras generated by the diadic sets of length $1/2^n$. Then define

$$X_n = 2^n \mathbf{1}[\omega \in [0, 2^{-n}]].$$

It is easy to check it is a martingale. Furthermore,

$$\int_{[0,1]} \xi dx = \sum_{j=0}^{\infty} \int_{[2^{-j-1}, 2^j]} X_j dx = \sum_{j=0}^{\infty} 1/2 = \infty.$$

□

Exercise 1.2 (5.2)**.** If X_n is a martingale such that the differences $Y_n = X_n - X_{n-1}$ are all square integrable, show that for $n \neq m$, $\mathbb{E}[Y_n Y_m] = 0$. Therefore

$$\mathbb{E}[X_n^2] = \mathbb{E}[X_0^2] + \sum_{j=1}^{n} \mathbb{E}[Y_j^2].$$

If in addition, $\sup_n \mathbb{E}[X_n^2] < \infty$, then show there is a random variable X such that

$$\lim_{n\to\infty} \mathbb{E}[|X - X_n|^2] = 0$$

Proof. Without loss of generality, suppose $n > m$. Then, by simply conditioning on $\mathcal{F}_{n-1}$

$$\mathbb{E}[Y_n Y_m] = \mathbb{E}[(X_n - X_{n-1})Y_m] = \mathbb{E}[Y_m(\mathbb{E}[X_n|\mathcal{F}_{n-1}] - X_{n-1})] = 0.$$

The other equality becomes obvious from the fact that $X_n = X_0 + \sum_{j=1}^n Y_j$. For the conclusion, note that together with the previous observation, it implies

$$\sum_n \mathbb{E}Y_n^2 < \infty$$

so that the X_n form a Cauchy sequence in L^2. As L^2 is complete, there exists such X. □

Exercise 1.3 (5.3)**.** Show there exists a uniformly bounded in L^1 martingale X_n but X_n does not converge in L^1.

Proof. The first exercise provides an example. It is clear that $X_n \to 0$ almost surely, but $|X_n|_{L^1} = 1$. This is an absurd, since any limit of X_n in L^1 would have to be a.s equal to $\lim_{n\to\infty} X_n$. □

Exercise 1.4 (5.4)**.** Estabilish that a nonnegative martingale has an almost sure limit (which may not be in L^1). Show (I wont really do it) that we can assume, wlog that we are in the following situation:

$$\Omega = \bigotimes_{j=1}^{\infty} \mathbb{R}, \quad \mathcal{F} = \sigma(x_1, \ldots, x_n), \quad X_j(\omega) = x_j.$$

The existence of a Q such that

$$\frac{dQ}{dP}|_{\mathcal{F}_n} = x_n$$

is essentially a Kolgomorov's consistency theorem.

Proof. Actually don't know how to do this one. □

Exercise 1.5 (5.5)**.** Show that $\tau(\omega) \equiv k$ is a stopping time for any constant k.

Proof. Note that if $n < k$, then $\{\tau \leqslant n\} = \varnothing \in \mathcal{F}_n$. Similarly, if $n \geqslant k$, $\{\tau \leqslant n\} = \Omega \in \mathcal{F}_n$. □

Exercise 1.6 (5.6)**.** Show that if τ is a stopping time and $f : \mathbb{N} \to \mathbb{N}$ is a non-decreasing function with $f(t) \geqslant t$ for all t. Then $f(\tau)$ is also a stopping time.

Proof. Just note that given n, there exists a maximal m s.t $f(m) \leqslant n$ and by hypothesis, $m \leqslant n$. Then $\{f(\tau) \leqslant n\} = \{\tau \leqslant m\} \in \mathcal{F}_m \subset \mathcal{F}_n$. □

Exercise 1.7 (5.7)**.** Show that if τ_1 and τ_2 are stopping times, so is $\max(\tau_1, \tau_2)$ and $\min(\tau_1, \tau_2)$. In particular, τ is an increasing limit of bounded stopping times $\tau \wedge n$.

Proof. Fix n, note that $\{\tau_1 \wedge \tau_2 \leqslant n\} = \{\tau_1 \leqslant n\} \cup \{\tau_1 \leqslant n\} \in \mathcal{F}_n$. The same is true for the max, we just take intersection instead of union. □

Exercise 1.8 (5.8)**.** Verify that for a stopping time τ, $\mathcal{F}_\tau$ is a sub σ-field. If $\tau \equiv k$, then $\mathcal{F}_\tau = \mathcal{F}_k$. If $\tau_1 \leqslant \tau_2$ are stopping times, then $\mathcal{F}_{\tau_1} \subset \mathcal{F}_{\tau_2}$. Finally, if τ is a stopping time, then it is $\mathcal{F}_\tau$ measurable.

Proof. Let's show that $\mathcal{F}_\tau \subset \mathcal{F}$ is a σ-field. Clearly $\varnothing \in \mathcal{F}_{\{\tau\}}$. Let $A \in \mathcal{F}_\tau$, then for every n, $A \cap \{\tau \leqslant n\} \in \mathcal{F}_n$, But then, $A^c \cup \{\tau > n\} \in \mathcal{F}_n$, and therefore $A^c \cap \{\tau \leqslant n\} \in \mathcal{F}_n$. Similarly, if $(A_n)_{n=1}^\infty$ is a sequence of sets in $\mathcal{F}_\tau$, then for any m

$$\{\tau \leqslant m\} \cap \bigcup_{n \geqslant 1} A_n = \bigcup_{n \geqslant 1} \{\tau \leqslant m\} \cap A_n \in \mathcal{F}_m.$$

If $\tau \equiv k$. It is clear that $\mathcal{F}_k \subset \mathcal{F}_\tau$ since for any $A \in \mathcal{F}_k$, $A \cap \{\tau \leqslant k\} = A \in \mathcal{F}_k$. Similarly, if $A \in \mathcal{F}_\tau$, then $A = A \cap \tau = k \in \mathcal{F}_k$.

If $\tau_1 \leqslant \tau_2$ and $A \in \mathcal{F}_{\tau_1}$ then for any m $A \cap \{\tau_2 \leqslant m\} \subset (A \cap \{\tau_1 \leqslant m\}) \cap \{\tau_2 \leqslant m\}$. Note that $A \cap \{\tau_1 \leqslant m\} \in \mathcal{F}_m$ and as $\{\tau \leqslant m\} \in \mathcal{F}_m$, we have that their intersection is also in $\mathcal{F}_m$.

Finally, measurability is obvious since $\{\tau \leqslant n\} \in \mathcal{F}_\tau$ for every n. □

Exercise 1.9 (5.10)**.** If X_n is a submartingale and $\tau_1 \leqslant \tau_2 \leqslant M$ are bounded stopping times, we have that $\mathbb{E}[X_{\tau_2}|\tau_1] \geqslant X_{\tau_1}$. With the respective result for supermartingales.

Proof. Decompose X_n as $Y_n + A_n$ where Y_n is a martingale and A_n is strictly increasing and predictable. Now it is clear that, by linearity,

$$\mathbb{E}[Y_{\tau_2} + A_{\tau_2}|\mathcal{F}_{\tau_1}] = Y_{\tau_1} + \mathbb{E}[A_{\tau_2}|\mathcal{F}_{\tau_1}] \geqslant Y_{\tau_1} + A_{\tau_1}.$$

since $A_{\tau_2} \geqslant A_{\tau_1}$.

The interesting case is when X_n is a martingale. Then you may only assume $\tau \leqslant M$ since

$$\mathbb{E}[X_m|\mathcal{F}_{\tau_1}] = \mathbb{E}[\mathbb{E}[X_m|\mathcal{F}_{\tau_2}]|\mathcal{F}_{\tau_1}].$$

Then we are left with proving that $\mathbb{E}[X_M|\mathcal{F}_\tau] = X_\tau$, to do that just separate on the values of τ. □

Exercise 1.10 (5.11)**.** Let $X_0 = 0$ and $X_n = \sum_{j \leqslant n} \xi_j$ where ξ_j are u. i.i.d on $\{-1, 1\}$. Let $\tau = \inf\{n : X_n = 1\}$. Then τ is a stopping time, $\mathbb{P}(\tau < \infty) = 1$, but τ is not bounded. $X_\tau = 1$ with probability 1 but $\mathbb{E}[X_0] = 0$.

Proof. I think the only thing to show on this exercise is that τ is a stopping time and $\mathbb{P}(\tau < \infty) = 1$. Note that $\{\tau = n\} = \{X_n = 1\} \cap \bigcap_{j \leqslant n} \{X_j \neq 1\} \in \mathcal{F}_n$, so it is a stopping time. To show reccurence we can proceed with the usual Catalan counting or use markov chains arguments. □

Exercise 1.11 (5.12)**.** Let τ be an unbounded stopping time. For every n, $\tau \wedge n$ is a bounded stopping time and $\mathbb{E}X_{\tau \wedge n} = 0$. As $n \to \infty$, $\tau \wedge n \to \tau$. If we can estabilish uniform integrability, then we can pass to the limit. In particular, is $S(\omega) = \sup_{0 \leqslant n \leqslant \tau} |X_n(\omega)|$ is integrable, then $\sup_n |X_{\tau \wedge n}| < S$ and so $\mathbb{E}X_\tau = 0$.

Proof. Yes, if Z_n converges almost surely to Z and they are uniformly integrable, then $\mathbb{E}Z_n \to \mathbb{E}Z$. The way to prove this is with Egorov's theorem. If μ is a finite measure and $Z_n \to Z$ a.s, then for every ε there exists a set A with $\mu(A^c) < \varepsilon$ such that $Z_n \to Z$ uniformly in A. By using uniform integrability we can win over in A^c and get usual convergence. Note that if S is integrable, then for all $X_{\tau\wedge n} \ll S$, so the family is uniformly integrable. And we can apply the previous result. □

Exercise 1.12 (5.14)**.** Seja τ um stopping time e $\tau_k \to \tau$ uma sequência crescente de stopping times. Mostre que

$$\mathcal{F}_\tau = \sigma\Big(\bigcup_k \mathcal{F}_{\tau_k}\Big)$$

Precisamos de algumas observações simples.

Lemma 1.1. Sejam $\tau_1 \leqslant \tau_2$ stopping times. Então $\mathcal{F}_{\tau_1} \subset \mathcal{F}_{\tau_2}$.

Proof. Tome $A \in \mathcal{F}_{\tau_1}$. Por definição, para todo n temos $A \cap \{\tau_1 \leqslant n\} \in \mathcal{F}_n$. Como $\tau_1 \leqslant \tau_2$ e $\{\tau_2 \leqslant n\} \in \mathcal{F}_n$,

$$A \cap \{\tau_2 \leqslant n\} = (A \cap \{\tau_1 \leqslant n\}) \cap \{\tau_2 \leqslant n\} \in \mathcal{F}_n.$$

Como isso vale para todo n, $A \in \mathcal{F}_{\tau_2}$. □

Observation 1.2. Seja $B \in \mathcal{F}_n$ e τ um stopping time qualquer. Então vale que

$$B \cap \{\tau = n\} \in \mathcal{F}_\tau \quad \& \quad B \cap \{\tau > n\} \in \mathcal{F}_\tau.$$

Prova do Exercício. Segue imediatamente do primeiro lema que $\sigma(\bigcup_k \mathcal{F}_{\tau_k}) \subset \mathcal{F}_\tau$. Basta verificar a outra inclusão. Seja $A \in \mathcal{F}_\tau$, separamos A de acordo com o que acontece em τ

$$A = A \cap \{\tau = \infty\} \cup \Big(\bigcup_n A \cap \{\tau \leqslant n\}\Big).$$

Vamos provar que cada pedacinho pertence a $\sigma(\bigcup_k \mathcal{F}_{\tau_k})$.

Por definição, $A \cap \{\tau \leqslant n\} \in \mathcal{F}_n$. Como $\tau_k \to \tau$, segue que

$$A \cap \{\tau \leqslant n\} = \bigcup_k [A \cap \{\tau \leqslant n\} \cap \{\tau_k = n\}].$$

Pela observação, anterior, $A \cap \{\tau \leqslant n\} \cap \{\tau_k = n\} \in \mathcal{F}_{\tau_k}$. Sendo união de conjuntos dessa forma, $A \cap \{\tau \leqslant n\} \in \sigma(\bigcup_k \mathcal{F}_{\tau_k})$. Para $\{\tau = \infty\}$ temos basicamente a mesma coisa. Note que

$$A \cap \{\tau = \infty\} = \bigcap_n A \cap \{\tau > n\}.$$

Usando que $\tau_k \to \tau$, $A \cap \{\tau > n\} = \bigcup_k A \cap \{\tau > n\} \cap \{t_k > n\}$. Da mesma forma que antes, cada um dos termos pertence a $\mathcal{F}_{\tau_k}$ e o resultado segue. □

Exercise 1.13. Let $X_n = \sum_{j=1}^n \xi_j$ where each ξ_j is a ± 1 with probability $1/2$. Show that if τ is a stopping time with $\mathbb{E}\tau < \infty$, then $\sup_{1\leqslant n\leqslant\tau} |X_n(\omega)|$ is square integrable and therefore $\mathbb{E}[X_\tau] = 0$. Hint: use the fact that $X_n^2 - n$ is a martingale.

Proof. So, first we need to check $X_n^2 - n$ is a martingale, but this follows easily from a previous calculation and the fact that $\mathbb{E}\xi_i\xi_j = 0$ if $i \neq j$. Now let's use the previous exercise, note that considering $\tau \wedge n$ as a bounded stopping time, we get that $\mathbb{E}X^2_{\tau\wedge n} - \tau \wedge n = 0$. So $\mathbb{E}X^2_{\tau\wedge n} \leqslant \mathbb{E}\tau \wedge n \leqslant \mathbb{E}\tau$. It follows that $\mathbb{E}X_{\tau\wedge n}$ is uniformly bounded in L^2 and therefore uniformly integrable. So $0 = \lim_{n\to\infty} \mathbb{E}X_{\tau\wedge n} = \mathbb{E}X_\tau$. □

Exercise 1.14. Show Doob's upcrossing inequality works for supermartingales, and chaging signs we get a downcrossing inequality for submartingales.

Proof. The only thing we need to check is that for supermartingales, $\mathbb{E}D(\omega) \leqslant 0$. All the other inequalities are the same. By switching signs we get the result for submartingales. □

Exercise 1.15. Let X_n be a symmetric random walk on $\mathbb{Z}$ starting from 0. Varadhan showed that for a fixed R and λ such that $\cos\lambda x \geqslant \cos\lambda R > 0$ for $x \in [-R, R]$ (we may take $\lambda \leqslant \pi/(2R)$) we have the inequality

$$\mathbb{E}_x[e^{\sigma\tau_R}] \leqslant \frac{\cos\lambda x}{\cos\lambda R}$$

where τ_R is the hitting time of $(-\infty, -R] \cup [R, \infty)$ and $\sigma = -\log\cos\lambda$. Can you show inequality above? What range is it valid in? Is $\mathbb{E}e^{\sigma\tau} < \infty$ for all σ?

Proof. The way Varadhan proves this is by considering the martingale $e^{\sigma n}\cos(\lambda X_n)$. Now it is clear that for every n,

$$\cos(\lambda x) = \mathbb{E}_x e^{\sigma\tau_R\wedge n}\cos(\lambda X_{n\wedge\tau}) = \mathbb{E}_x[\mathbf{1}[\tau > n]e^{\sigma n}\cos(\lambda X_n)] + \mathbb{E}_x[\mathbf{1}[\tau \leqslant n]e^{\sigma\tau}\cos(\lambda R)].$$

If $\sigma < -\log\cos\lambda$, then the first term of the right hand side is bounded by $\mathbb{E}e^{\sigma n}\cos(\lambda X_n) \to 0$ when $n \to \infty$. As we know the second term converges to $\mathbb{E}e^{\sigma\tau}\cos(\lambda R)$ we get equality in this case. I should work a bit harder on the $\sigma = -\log\cos\lambda$ case.

For the last observation, note that trivially $\mathbb{P}(X_1 = 1, X_2 = 0, \ldots X_n = n\%2) = 2^{-n}$. So $\mathbb{E}e^{\sigma\tau} \geqslant 2^{-n}e^{\sigma n}$, in particular if $\sigma > \log 2$, as this is valid for all n, we already get $\mathbb{E}e^{\sigma\tau} = \infty$. □

2 Durrett

Alguns exercícios do Durrett.

2.1 Martingales

Exercise 2.1 (4.2.4)**.** Let $\{X_n : n \geqslant 0\}$ be a submartingale with $\sup X_n < M_1 < \infty$. Let $\xi_n = X_n - X_{n-1}$ and suppose $\mathbb{E}\sup\xi_n^+ < M_2 < \infty$. Show that X_n converges almost surely.

Proof. I don't get why we need the ξ_n hypothesis. $X_n^+ \leqslant 0 \vee M_1$. So $\mathbb{E}X_n^+ \leqslant 0 \vee M_1$ and by the bounded submartingale convergence theorem, we converge a.s. □

Exercise 2.2 (4.2.5)**.** Give an example of a martingale X_n with $X_n \to -\infty$ a.s. Hint let $X_n = \xi_1 + \ldots \xi_n$ where the ξ_j are independent.

Proof. Consider ξ_m with $\mathbb{E}\xi_m = 0$ s.t

$$\mathbb{P}(\xi_m = -1) = 1 - 2^{-m} \quad \& \quad \mathbb{P}(\xi_m = 2^m(1 - 2^{-m})) = 2^{-m}.$$

It is clear that X_n forms a martingale, but by Borel Cantelli, $\xi_m \neq -1$ for only finitely many m almost surely. So $X_n \to -\infty$ a.s. □

Exercise 2.3 (4.2.6)**.** Let $Y_1, \dots$ be non negative i.i.d random variables with $\mathbb{E}Y_m = 1$ and $\mathbb{P}(Y_m = 1) < 1$. $X_n = \prod_{m \leqslant n} Y_m$ defines a martingale. Show that $X_n \to 0$ a.s and that $\frac{1}{n} \log X_n \to c$ a.s for some $c < 0$.

Proof. Note that X_m is a nonnegative inferiorly bounded martingale, therefore a supermartingale and we know X_m converges a.s. We also know that for some $\alpha < 1$, $Y_m < \alpha$ infinitely often with probability 1 by the second Borel Cantelli. So X_m must converge to 0. The part follows trivially by the law of large numbers and that $X_n \to 0$ a.s. □

Exercise 2.4 (4.2.8)**.** Let X_n and Y_n be positive integrable and adapted to $\mathcal{F}_n$. Suppose

$$\mathbb{E}[X_{n+1}|\mathcal{F}] \leqslant (1 + Y_n)X_n$$

with $\sum Y_n < \infty$ a.s. Prove that X_n converges almost surely to a finite limit by findind an appropriate supermartingale.

Proof. Consider

$$Z_n = \mathbb{E}[X_n \prod_{m \geqslant n} (1 + Y_m)|\mathcal{F}_n]$$

which is integrable and positive by assumption. Now by the tower property,

$$\mathbb{E}[Z_{n+1}|\mathcal{F}_n] = \mathbb{E}[\mathbb{E}[X_{n+1} \prod_{m \geqslant n+1} (1 + Y_m)|\mathcal{F}_{n+1}]|\mathcal{F}_n].$$

Furthermore, this right hand side is equal to

$$\mathbb{E}[X_{n+1}\mathbb{E}[\prod_{m \geqslant n+1|\mathcal{F}_{n+1}}]|\mathcal{F}_n] \leqslant \mathbb{E}[X_n(1 + Y_n)\mathbb{E}[\prod_{m \geqslant n+1} (1 + Y_m)|\mathcal{F}_{n+1}]|\mathcal{F}_n] \leqslant Z_n.$$

So by the Martingale Convergence Theorem, Z_n converges a.s. But, as $\sum_m Y_m < \infty$ a.s, $Z_n \to X_n$, so X_n converges too. □

Exercise 2.5 (4.2.9)**.** Switching principle. Suppose X_n^1 and X_n^2 are supermartingales with respect to $\mathcal{F}_n$ and N is a stopping time so that $X_N^1 \geqslant X_N^2$. Then both

$$Y_n = X_n^1 \mathbf{1}[N > n] + X_n^2 \mathbf{1}[N \leqslant n] \quad \& \quad Z_n = X_n^1 \mathbf{1}[N \geqslant n] + X_n^2 \mathbf{1}[N < n]$$

are supermartingales.

Proof. Let's do the first one. As always we separate the as the possible value of N. Clearly Y_n and Z_n are $\mathcal{F}_n$ measurable. Note that

$$\mathbb{E}[Y_n|\mathcal{F}_{n-1}] = \mathbb{E}[X_n^1\mathbf{1}[N > n]|\mathcal{F}_{n-1}] + \mathbb{E}[X_n^2\mathbf{1}[N = n]|\mathcal{F}_{n-1}] + \mathbb{E}[X_n^2\mathbf{1}[N < n]|\mathcal{F}_{n-1}].$$

Now we exchange and separate the measurable variables to get that the righthand side is less than

$$\mathbf{1}[N > n-1]\mathbb{E}[X_n^1|\mathcal{F}_{n-1}] + \mathbf{1}[N < n]\mathbb{E}[X_n^2|\mathcal{F}_{n-1}] \leqslant Y_{n-1}$$

because both X_n^1, X_n^2 are supermartingales. Let's see the same thing for Z_n.

$$\mathbb{E}[Z_n|\mathcal{F}_{n-1}] = \mathbf{1}[N > n-1]\mathbb{E}[X_n^1|\mathcal{F}_{n-1}] + \mathbb{E}[\mathbf{1}[N = n-1]X_N^2|\mathcal{F}_{n-1}] + \mathbf{1}[N < n-1]\mathbb{E}[X_n^2|\mathcal{F}_{n-1}] \leqslant Z_{n-1}.$$

by the same argument as before. □

2.2 Markov Chains

Exercise 2.6 (5.2.1)**.** Let $A \in \sigma(X_0, \dots, X_n)$ and $B \in \sigma(X_n, \dots)$. Use the Markov property to show that for any initial distribution μ

$$\mathbb{P}_\mu(A \cap B|X_n) = \mathbb{P}_\mu(A|X_n)\mathbb{P}_\mu(B|X_n).$$

Proof. We write it out using conditional expectations,

$$\mathbb{P}_\mu(A \cap B|X_n) = \mathbb{E}_\mu[\mathbf{1}[A]\mathbb{E}_\mu[\mathbf{1}[B]|\mathcal{F}_n]|X_n] = \mathbb{E}_\mu[\mathbf{1}[A]\mathbb{P}_\mu(B|X_n)|X_n] = \mathbb{P}_\mu(A|X_n)\mathbb{P}_\mu(B|X_n).$$

□

Exercise 2.7 (5.2.2)**.** Let X_n be a Markov chain. Use Lévy's 0-1 law to show that if

$$\mathbb{P}(\bigcup_{m \geqslant n} \{X_m \in B_m\}|X_n) \geqslant \delta > 0 \quad \text{on } \{X_n \in A_n\}$$

then $\mathbb{P}(\{X_n \in A_n \, i.o\} - \{X_n \in B_n \, i.o\}) = 0$.

Proof. Let $Z = \mathbf{1}[X_n \in B_n \, i.o]$. By the Markov property, $Z_n = \mathbb{E}[Z|\mathcal{F}_n] = \mathbb{E}[Z|X_n]$ is a unif. bounded martingale and therefore converges a.s. to 0 or 1. On $\{X_n \in A_n \, i.o\}$, $Z_n > \delta i.o$ so Z_n must converge to $Z = 1$. It follows that $\mathbf{1}[X_n \in A_n \, i.o] = \mathbf{1}[X_n in A_n \, i.o \cap X_n \in B_n \, i.o]$. □

Exercise 2.8 (5.2.3)**.** A state a is called absorbing if $\mathbb{P}_a(X_1 = a) = 1$. Let $D = \{\exists n \geqslant 1, X_n = a\}$ and let $h(x) = \mathbb{P}_x(D)$. Conclude that $h(X_n) \to 0$ a.s on D^c.

Proof. Note that, by the Markov property, $D = \{X_n = a \, i.o\}$. By Levy's, $\mathbb{P}_{X_n}(D) = \mathbb{E}[\mathbf{1}[D]|\mathcal{F}_n] \to \mathbf{1}[D]$ a.s. So on D^c we have the result. □

Exercise 2.9 (5.2.4)**.** Let $\tau_y = \inf\{n \geqslant 1 : X_n = y\}$. Show that

$$p^n(x, y) = \sum_{m=1}^{n} \mathbb{P}_x(T_y = m)p^{n-m}(y, y).$$

Proof. Consider $\tau = \tau_y$, then

$$p^n(x,y) = \mathbb{P}_x(X_n = y) = \mathbb{E}_x[\mathbf{1}[\tau \leqslant n]\mathbb{E}[\mathbf{1}[X_n = y]|\mathcal{F}_\tau]]$$

By the strong Markov property, that is the same as

$$\mathbb{E}_x\mathbf{1}[\tau \leqslant n]\mathbb{E}_{X_\tau}\mathbf{1}[X_{n-\tau} = y] = \sum_{m=1} \mathbb{P}_x(\tau = m)p^{n-m}(y,y).$$

□

Exercise 2.10 (5.2.5)**.** Show that

$$\sum_{m=0}^{n} \mathbb{P}_x(X_m = x) \geqslant \sum_{m=k}^{n+k} \mathbb{P}_x(X_m = x).$$

Proof. Ok, I remember I did this one but got stuck on something stupid. First thing, if we define $h_n(z) = \sum_{m=0}^n \mathbb{P}_z(X_m = x)$, we show inductively that $h_n(z) \leqslant h_n(x)$ for all z. It is clearly true for h_0. Now note that if $z \neq x$,

$$h(z) = \sum_{m=1}^{n} \mathbb{P}_z(X_m = x) = \mathbb{E}_z h_{n-1}(X_1) \leqslant h_{n-1}(x) \leqslant h_n(x).$$

But then it is clear that

$$\sum_{m=0}^{n} \mathbb{P}_x(X_{m+k} = x) = \mathbb{E}_x \sum_{m=0}^{n} \mathbb{P}_{X_k}(X_m = x) = \mathbb{E}_x h_n(X_k) \leqslant h_n(x).$$

□

Exercise 2.11 (5.2.6)**.** Let $\tau_C = \inf\{n \geqslant 1 : X_n \in C\}$. Suppose $S - C$ finite and for each $x \in S - C$ $\mathbb{P}_x(\tau_C < \infty) > 0$. Then there is as $N < \infty$ and $\varepsilon > 0$ so that for all $y \in S - C$, $\mathbb{P}_y(\tau_C > kN) \leqslant (1-\varepsilon)^k$.

Proof. First note that, because $S - C$ is finite, there is ε and N such that $\mathbb{P}_x(\tau_C < N) > \varepsilon$ for all $x \in S - C$. Now we prove the result by induction, clearly, $\mathbb{P}_y(\tau_C > 0) \leqslant 1$ for $y \in S - C$. It follows from the weak Markov property that $\mathbb{P}_y(\tau_C > (k+1)N)$ is equal to

$$\mathbb{E}[\mathbf{1}[\tau_C > kN]\mathbb{E}[\mathbf{1}[\tau_C > (k+1)N]|\mathcal{F}_{kN}]] = \mathbb{E}[\mathbf{1}[\tau_C > kN]\mathbb{P}_{X_{kN}}(\tau_C > N)] \leqslant (1-\varepsilon)^{k+1}.$$

□

Exercise 2.12 (5.2.7)**.** Let $V_C = \inf\{n \geqslant 0 : X_n \in C\}$ and let $h(x) = \mathbb{P}_x(V_A < V_B)$. Suppose $A \cap B = \varnothing$, $S - (A \cup B)$ finite and $\mathbb{P}_x(V_{A\cup B} < \infty) > 0$ for all $x \notin A \cup B$. Show that

$$h(x) = \sum_y p(x,y)h(y) \quad \text{for } x \notin A \cup B.$$

That if some h satisfies that, then $h(X_{n\cap V_{A\cup B}})$ is a martingale. Conclude that if h satisfies that, then $\mathbb{P}_x(V_A < V_B)$ is the only solution that satisfies $h = 1$ on A and $h = 0$ on B.

Proof. Note that if $x \notin A \cup B$, then

$$\mathbb{P}_x(V_A < V_B) = \mathbb{E}_x[\mathbb{P}_{X_1}(V_A < V_B)] = \sum_y p(x,y)h(y).$$

Suppose that $h(x)$ satisfies that relation and $|h(x)| < \infty$ for $x \notin A \cup B$. Note that by the previous exercises, $V_{A\cup B} < \infty$ a.s. It is clear that if h is integrable, then $h(X_{n\wedge V_{A\cup B}})$ is a martingale, since, setting $\tau = V_{A\cup B}$,

$$\mathbb{E}[h(X_{n\wedge\tau})|\mathcal{F}_{n-1}] = \sum_{j=0}^{n-1} \mathbf{1}[\tau = j]h(X_j) + \mathbb{E}_{X_{n-1}}[\mathbf{1}[\tau \geqslant 1]h(X_1)] = h(X_{\tau\wedge(n-1)}).$$

To show that it is integrable we proceed by induction on n. Clearly, for $X_0 \in S-C$, $|h(X_{\tau\wedge 0})| < \infty$. Now note that

$$\mathbb{E}[|h(X_{\tau\wedge n+1})|] = \mathbb{E}[\mathbf{1}[\tau \geqslant n+1]|h(X_{n+1})| + \mathbf{1}[\tau \leqslant n]|h(X_\tau)|]$$

Now, by assumption $\mathbb{E}[\mathbf{1}[\tau \leqslant n]||h(X_\tau)| + |h(X_n)|\mathbf{1}[\tau > n]] < \infty$. So second term from the previous sum has to be finite. We are left with proving that $\mathbb{E}[\mathbf{1}[\tau \geqslant n+1]|h(X_{n+1})|] < \infty$. But for this we expand and use the property.

$$\mathbb{E}_{x_0}[\mathbf{1}[\tau \geqslant n+1]|h(X_{n+1})|] \leqslant \sup_{y\in S-C} \mathbb{E}_y[|h(X_1)|] < \infty.$$

We conclude that it is a martingale. Note that, by hypothesis, if it is 1 on A and 0 on B it is uniformly bounded in L^∞, so it converges a.s to what we want. □

Exercise 2.13 (5.2.8)**.** Let X_n be a Markov chain with $S = \{0, 1, \ldots, N\}$ and suppose that X_n is a martingale. Let $\tau_x = \inf\{n \geqslant 0 : X_n = x\}$ and suppose that $\mathbb{P}_x(\tau_0 \wedge \tau_N < \infty) > 0$ for all x. Show that $\mathbb{P}_x(\tau_N < \tau_0) = x/N$.

Proof. We've seen that almost surely, $\tau = \tau_0 \wedge \tau_N < \infty$. Now, it is clear that, since the space is finite, $X_{n\wedge\tau}$ is a uniformly integrable martingale that converges to X_τ. We thus conclude that

$$x = \mathbb{E}_x X_\tau = \mathbb{E}_x[\lim_{n\to\infty} X_{\tau\wedge n}] = N\mathbb{P}_x(\tau_N < \tau_0).$$

□

Exercise 2.14 (5.2.11)**.** Let $\tau_A = \inf\{n \geqslant 0 : X_n \in A\}$ and $g(x) = \mathbb{E}_x\tau_A$. Suppose that $S - A$ is finite and for each $x \in S - A$, $\mathbb{P}_x(\tau_A < \infty) > 0$. Show that

$$g(x) = 1 + \sum_y p(x,y)g(y) \quad \text{for } x \notin A.$$

Show that if g satisfies the previous equation, then $g(X_{n\wedge\tau_A}) + n \wedge \tau_A$ is a martingale. Conclude that $\mathbb{E}_x\tau_A$ is the only solution that is 0 on A.

Proof. This one is very similar to exercise 5.2.7. Note that by the weak Markov property, for $x \notin A$

$$\mathbb{E}_x[\tau_A] = \mathbb{E}_x\mathbb{E}[\tau_A|\mathcal{F}_1] = 1 + \mathbb{E}_x[\mathbb{E}_{X_1}\tau_A].$$

Just as before, $Z_n = g(X_{n\wedge\tau_A}) + n \wedge \tau_A$ is integrable when we assume $\mathbb{E}_y p(x,y)|g(y)| < \infty$ for all x outside A. And it follows that

$$\mathbb{E}[Z_n|\mathcal{F}_{n-1}] = \sum_{j=0}^{n-1} \mathbf{1}[\tau = j](g(X_j) + j) + \mathbf{1}[\tau \geqslant n]\mathbb{E}[g(X_n) + 1 + (n-1)|\mathcal{F}_{n-1}]$$

and this righthand side is

$$\mathbf{1}[\tau \leqslant n-1](g(X_\tau) + \tau) + \mathbf{1}[\tau \geqslant n](g(X_{n-1}) + n - 1) = Z_{n-1}.$$

For the last bit, note that if $g = 0$ on A, then we are uniformly integrable. By a previous exercise, $\tau < \infty$ a.s. So Z_n converges almost surely and for $x \notin A$,

$$g(x) = \mathbb{E}_x[Z_0] = \mathbb{E}_x[\lim_{n\to\infty} Z_n] = \mathbb{E}_x[\tau_A].$$

□

For the next few exercises we make the following definition.

Definition 2.1. If (S,p) is a Markov chain and $x, y \in S$, then we set $\rho(x,y) = \mathbb{P}_x(\tau_y < \infty)$.

Exercise 2.15 (5.3.1)**.** Suppose y is recurrent and for $k \geqslant 0$ let $R_k = T_y^k$ be the time of the kth return to y, and for $k \geqslant 1$ let $r_k = R_k - R_{k-1}$ be the kth interarrival time. Use the strong Markov property to conclude that under $\mathbb{P}_y$, the vectors $v_k = (r_k, X_{R_{k-1}}, \dots, X_{R_k-1})$ are i.i.d.

Proof. A stronger thing follows, the distribution of $(X_{R_k+1}, \dots)$ is independent of the previous interarrival times. To this end, note that $R_k < \infty$ a.s, and by the Strong Markov Property, for every measurable Y on Ω_0,

$$\mathbb{E}_y[Y \circ \Theta_{R_k}|\mathcal{F}_{R_k}] = \mathbb{E}_y[Y].$$

In particular, for measurable Y_k dependent on $(X_{R_{k-1}}, \dots X_{R_{k-1}})$ we find that

$$\mathbb{E}_y[\prod_{j=1}^{n} Y_j \circ \Theta_{R_{j-1}}] = \mathbb{E}_y[\prod_{j=1}^{n-1} Y_j \Theta_{R_j}\mathbb{E}[Y_n \circ \Theta_{R_{n-1}}|\mathcal{F}_{R_{n-1}}]] = \mathbb{E}_y[Y_n]\mathbb{E}_y[\prod_{j=1}^{n-1} Y_j \circ \Theta_{R_{j-1}}].$$

Applying this for Y_j being indicators, we conclude that these distributions are i.i.d. □

Exercise 2.16 (5.3.2)**.** Use the strong Markov property to show that $\rho_{xz} \geqslant \rho_{x,y}\rho_{yz}$.

Proof. Let τ_z and τ_y be defined as usual. We want to show that

$$\mathbb{P}_x(\tau_z < \infty) \geqslant \mathbb{P}_x(\tau_y < \infty)\mathbb{P}_y(\tau_z < \infty).$$

Note that,

$$\mathbb{P}_x(\tau_z < \infty) \geqslant \mathbb{E}_x[\mathbf{1}[\tau_y < \infty]\mathbb{P}_x(\tau_z < \infty)] = \mathbb{E}_x\mathbf{1}[\tau_y < \infty]\mathbb{E}_x[\mathbb{E}[\mathbf{1}[\tau_z < \infty]|\mathcal{F}_{\tau_y}]]$$

Now this right hand side is equal to

$$\mathbb{E}_x\mathbf{1}[\tau_y < \infty]\mathbb{P}_y(\tau_z < \infty) = \rho_{x,y}\rho(y,z)$$

which is what we want to show. □

Exercise 2.17 (5.3.5)**.** Suppose S is irreducible, $\varphi > 0$ with $\mathbb{E}_x\varphi(X_1) \leqslant \varphi(x)$ for $x \notin F$, a finite set, and $\phi(x) \to 0$ as $x \to \infty$, then the chain is transient.

Proof. Consider $\tau = \inf n \geqslant 0 : X_n \in F$ and the positive supermartingale given by $Y_n = \varphi(X_{n\wedge\tau})$. It is a supermartingale as, using the weak Markov property,

$$\mathbb{E}[Y_{n+1}|\mathcal{F}_n] = \mathbf{1}[\tau \leqslant n]\varphi(X_\tau)+\mathbf{1}[\tau > n]\mathbb{E}[\varphi(X_{n+1})|\mathcal{F}_n] = \mathbf{1}[\tau \leqslant n]\varphi(X_\tau)+\mathbf{1}[\tau > n]\mathbb{E}_{X_n}\varphi(X_1) \leqslant Y_n.$$

So, by the martingale convergence theorem, this converges a.s to a limit Z with $\mathbb{E}[Z] \leqslant \mathbb{E}_x\varphi(X_0)$. Furthermore, consider the stopping time $\tau_\varepsilon = \inf\{n \geqslant 0 : \varphi(X_n) \leqslant \varepsilon\}$. It is clear that $Z_{\{}n \wedge \tau_\varepsilon\}$ is also a supermartingale, and, because $\varphi(x) \to \infty$, $\tau_\varepsilon < \infty$ a.s. So we also converge with this positive supermartingale. But then, for $x \notin F$,

$$\varphi(x) = \mathbb{E}_x[\varphi(X_{\tau\wedge\tau_\varepsilon})] = \mathbb{E}\mathbf{1}[\tau < \tau_\varepsilon]\varphi(X_\tau) + \mathbb{E}\mathbf{1}[\tau_\varepsilon \leqslant \tau]\varphi(X_{\tau_\varepsilon}).$$

Rearranging we see that for every $\varepsilon > 0$,

$$\mathbb{P}(\{\tau < \tau_\varepsilon\}) \leqslant \frac{\varphi(x) + \varepsilon}{\inf\{\varphi(y) : y \in F\}}$$

Sending $x \to \infty$ we find x such that $\phi(x)/\inf\{\phi(y) : y \in F\} < 1/10$. Now

$$\mathbb{P}_x(\tau < \infty) = \mathbb{P}(\bigcup_{\varepsilon\to 0}\{\tau < \tau_\varepsilon\}) = \lim_{\varepsilon\to 0}\mathbb{P}(\tau < \tau_\varepsilon) \leqslant \frac{1}{10}.$$

proving transience. □

Exercise 2.18 (5.3.7)**.** f is said to be superharmonic if $\mathbb{E}_x(f(X_1)) \leqslant f(x)$ or equivalentely, $f(X_n)$ is a supermartingale. Suppose p is irreducibible, show that p is recurrent iff every non-negative superharmonic function is constant.

Proof. Suppose that every non-negative superharmonic function is constant and consider $h_z(x) = \mathbb{P}_x(\tau_z < \infty)$. Then $h_z(z) = 1$ but h is harmonic, as

$$\mathbb{P}_x(\tau_z < \infty) = \mathbb{E}_x[\mathbb{P}(\tau_z < \infty|X_1)] = \sum_y p(x,y)h_z(y).$$

therefore superharmonic and constant equal to 1.

On the other hand, suppose that p is recurrent. As $f(X_n)$ is a positive superharmonic martingale, it converges a.s. But X_n infinitely often is equal to any x of the chain, so f must be constant. □

Exercise 2.19 (5.5.2)**.** Let $w_{xy} = \mathbb{P}_x(T_y < T_x)$. Show that $\mu_x(y) = w_{xy}/w_{yx}$.

Exercise 2.20 (5.5.3)**.** Show that if p is irreducible and recurrent, then

$$\mu_x(y)\mu_y(z) = \mu_x(z)$$

Exercise 2.21 (5.5.4)**.** Suppose p is irreducible and positive reccurent. Then $\mathbb{E}_x T_y < \infty$ for all x, y.

Exercise 2.22 (5.5.5)**.** Suppose p is irreducible and has a stationary measure μ with $\mathbb{E}_x \mu(x) = \infty$. Then p is not positive recurrent.